STAT 110 – Introduction to Probability

Course Notes

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Chapter 2: Conditional Probability (Selected Sections)

2.8 Pitfalls and Paradoxes

Conditional probability is the mathematical framework for updating beliefs when presented with evidence. However, intuitive assessments of conditional evidence are notoriously error-prone. Two classic errors in legal reasoning are the **prosecutor's fallacy** and the **defense attorney's fallacy**. Another critical pitfall is **Simpson's paradox**, where trends observed within separate subgroups reverse when the data are pooled.

2.8.1 Prosecutor's Fallacy

Intuition 1.1. The prosecutor's fallacy is the confusion of $P(A \mid B)$ with $P(B \mid A)$. Specifically, it treats the probability of the observed evidence given innocence as if it were the probability of innocence given the evidence. Even if an innocent person is extraordinarily unlikely to match a piece of evidence, a matching suspect is not necessarily likely to be guilty.

Example 1.1 (The Sally Clark Case)**.** In 1998 in the United Kingdom, Sally Clark was tried for murder after two of her infant sons died suddenly shortly after birth. An expert medical witness for the prosecution testified that the probability of a newborn dying of Sudden Infant Death Syndrome (SIDS) was approximately 1/8500. The witness calculated the probability of two SIDS deaths within the same household as:

$$\left(\frac{1}{8500}\right)^2 \approx \frac{1}{72{,}250{,}000} \approx \frac{1}{73 \text{ million}}.$$

The expert then asserted that the probability of Clark's innocence was 1 in 73 million.

This argument contains two major statistical errors:

1. **Violation of Independence:** The calculation multiplies the individual probabilities together. This is only valid if SIDS deaths within a family are independent. In reality, shared genetics and household environments mean that if one child dies of SIDS, the risk to a subsequent sibling is substantially elevated.
2. **Transposition of Conditionals:** The witness equated the probability of the evidence given innocence with the probability of innocence given the evidence:

$$P(\text{evidence} \mid \text{innocence}) \neq P(\text{innocence} \mid \text{evidence}).$$

By Bayes' Rule, the posterior probability of innocence given the evidence is:

$$P(\text{innocence} \mid \text{evidence}) = \frac{P(\text{evidence} \mid \text{innocence})P(\text{innocence})}{P(\text{evidence})}.$$

Expanding the denominator with the Law of Total Probability:

$$P(\text{evidence}) = P(\text{evidence} \mid \text{innocence})P(\text{innocence}) + P(\text{evidence} \mid \text{guilt})P(\text{guilt}).$$

Although two SIDS deaths in a family are rare, double infanticides are also extremely rare ($P(\text{guilt}) \approx 0$). Consequently, the second term in the denominator does not dominate, and the posterior probability of innocence remains high. Sally Clark's conviction was later overturned.

2.8.2 Defense Attorney's Fallacy

Intuition 1.2. The defense attorney's fallacy occurs when an advocate fails to condition on the full body of available evidence. An unconditional event probability might seem small, but once conditioned on the fact that a crime has occurred, the same evidence can provide strong support for guilt.

Example 1.2 (Prior Abuse in a Murder Trial)**.** A woman is murdered, and her husband is tried for the crime. The prosecution presents evidence that the husband had a history of physically abusing his wife. The defense attorney argues that the history of abuse is irrelevant, stating that only 1 in 10,000 men who abuse their wives ultimately murder them.

Assume the defense attorney's figure is correct: $P(G \mid A) = 1/10{,}000$, where A denotes abuse and G denotes guilt. Assume also:

- 1 in 10 husbands abuse their wives: $P(A) = 0.10$.
- 1 in 5 murdered wives are killed by their husbands: $P(G \mid M) = 0.20$, where M is the event that the wife was murdered.
- 50% of husbands who murder their wives previously abused them: $P(A \mid G, M) = 0.50$.
- For an innocent husband, the probability of abuse equals the baseline rate: $P(A \mid G^c, M) = P(A) = 0.10$.

The Error: The defense attorney cites $P(G \mid A)$, but this ignores the fact that the wife has already been murdered. The correct quantity of interest is $P(G \mid A, M)$.

Applying Bayes' Rule with extra conditioning on M:

$$\begin{aligned} P(G \mid A, M) &= \frac{P(A \mid G, M)P(G \mid M)}{P(A \mid G, M)P(G \mid M) + P(A \mid G^c, M)P(G^c \mid M)} \\ &= \frac{(0.50)(0.20)}{(0.50)(0.20) + (0.10)(0.80)} \\ &= \frac{0.10}{0.10 + 0.08} \\ &= \frac{0.10}{0.18} = \frac{5}{9} \approx 0.556. \end{aligned}$$

Conditioning on the abuse elevates the probability of guilt from the prior baseline of 20% to over 55%.

2.8.3 Simpson's Paradox

Intuition 1.3. Simpson's paradox occurs when an apparent trend across several subgroups disappears or reverses when the groups are pooled. This happens when an unobserved confounding variable influences group assignment and the outcome simultaneously.

Example 1.3 (Dr. Hibbert vs. Dr. Nick)**.** Two doctors, Dr. Hibbert and Dr. Nick, perform heart surgeries and Band-Aid removals. Their records are shown below:

	Dr. Hibbert		**Dr. Nick**	
	Heart	Band-Aid	Heart	Band-Aid
Success	70	10	2	81
Failure	20	0	8	9
Rate	$\frac{70}{90} \approx 77.8\%$	$\frac{10}{10} = 100\%$	$\frac{2}{10} = 20.0\%$	$\frac{81}{90} = 90.0\%$
Combined	$\frac{80}{100} = 80.0\%$		$\frac{83}{100} = 83.0\%$	

Dr. Hibbert has a higher success rate for both procedures:

$$\text{Heart: } 77.8\% > 20.0\%, \qquad \text{Band-Aid: } 100\% > 90.0\%.$$

Yet in the aggregate, Dr. Nick appears to have the higher overall success rate ($83.0\% > 80.0\%$). Dr. Hibbert's combined rate is lower because 90% of his cases are high-risk heart surgeries, whereas 90% of Dr. Nick's cases are low-risk Band-Aid removals.

Mathematical Explanation via LOTP: Let A denote surgical success, B denote Dr. Nick (B^c is Dr. Hibbert), and C denote heart surgery (C^c is Band-Aid removal). Simpson's paradox occurs when:

$$P(A \mid B, C) < P(A \mid B^c, C) \quad \text{and} \quad P(A \mid B, C^c) < P(A \mid B^c, C^c),$$

yet

$$P(A \mid B) > P(A \mid B^c).$$

Expanding via the Law of Total Probability:

$$\begin{aligned} P(A \mid B) &= P(A \mid C, B)P(C \mid B) + P(A \mid C^c, B)P(C^c \mid B) \\ &= (0.20)(0.10) + (0.90)(0.90) = 0.02 + 0.81 = 0.83, \\ P(A \mid B^c) &= P(A \mid C, B^c)P(C \mid B^c) + P(A \mid C^c, B^c)P(C^c \mid B^c) \\ &= \left(\frac{70}{90}\right)(0.90) + (1.00)(0.10) = 0.70 + 0.10 = 0.80. \end{aligned}$$

The weights $P(C \mid B) = 0.10$ and $P(C \mid B^c) = 0.90$ are heavily skewed, causing the aggregate inequality to flip.

2.9 Recap

- **Conditional Probability:** For $P(B) > 0$, $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$.
- **Bayes' Rule:** $P(A \mid B) = \dfrac{P(B \mid A)P(A)}{P(B)}$.
- **Law of Total Probability (LOTP):** For a partition $A_1, \ldots, A_n$ of S,

$$P(B) = \sum_{i=1}^{n} P(B \mid A_i)P(A_i).$$

- **Problem-Solving Heuristic:** Condition on what you wish you knew.

Chapter 3: Random Variables and their Distributions

3.1 Random Variables

Definition 2.1 (Random Variable)**.** Given an experiment with sample space S, a **random variable** (r.v.) X is a function from the sample space S to the real line $\mathbb{R}$:

$$X : S \to \mathbb{R}.$$

Intuition 2.1. A random variable is a deterministic numerical summary of the outcome of a random experiment. The randomness comes entirely from the experiment choosing an outcome $s \in S$ according to a probability function P. Once s is realized, the random variable crystallizes into a definite real value $X(s)$.

Example 2.1 (Coin Tosses)**.** Flip a fair coin twice: $S = \{HH, HT, TH, TT\}$.

- Let X be the number of Heads: $X(HH) = 2$, $X(HT) = X(TH) = 1$, and $X(TT) = 0$.
- Let Y be the number of Tails: $Y = 2 - X$, meaning $Y(s) = 2 - X(s)$ for all $s \in S$.
- Let I be the indicator that the first toss is Heads: $I(HH) = I(HT) = 1$, and $I(TH) = I(TT) = 0$.

Encoding $H = 1$ and $T = 0$, the sample space is $S = \{(1,1), (1,0), (0,1), (0,0)\}$. The functions can be written as:

$$X(s_1, s_2) = s_1 + s_2, \quad Y(s_1, s_2) = 2 - s_1 - s_2, \quad I(s_1, s_2) = s_1.$$

3.2 Distributions and Probability Mass Functions

Definition 2.2 (Discrete Random Variable and Support)**.** An r.v. X is **discrete** if there exists a finite or countably infinite sequence of values $a_1, a_2, \dots$ such that $P(X = a_j \text{ for some } j) = 1$. The set of values x for which $P(X = x) > 0$ is called the **support** of X.

Definition 2.3 (Probability Mass Function)**.** The **Probability Mass Function** (PMF) of a discrete r.v. X is the function p_X given by:

$$p_X(x) = P(X = x).$$

Theorem 2.1 (Valid PMF Criteria)**.** *A function p_X is a valid PMF for a discrete r.v. X with support $\{x_1, x_2, \dots\}$ if and only if:*

1. ***Nonnegative:*** *$p_X(x) > 0$ if $x = x_j$ for some j, and $p_X(x) = 0$ otherwise.*
2. ***Sums to 1:*** $\sum_j p_X(x_j) = 1$.

Example 2.2 (PMFs of Two Coin Tosses)**.** For the two-toss experiment:

- $p_X(0) = \frac{1}{4}, \quad p_X(1) = \frac{1}{2}, \quad p_X(2) = \frac{1}{4}.$

- $p_Y(0) = \frac{1}{4}, \quad p_Y(1) = \frac{1}{2}, \quad p_Y(2) = \frac{1}{4}.$
- $p_I(0) = \frac{1}{2}, \quad p_I(1) = \frac{1}{2}.$

X and Y have identical PMFs ($p_X = p_Y$) even though they are different random variables.

Example 2.3 (Sum of Two Dice)**.** Roll two fair six-sided dice, producing independent rolls X and Y. Let $T = X + Y$. The sample space has 36 equally likely outcomes. The PMF of T is:

$$P(T = 2) = P(T = 12) = \frac{1}{36}, \qquad P(T = 5) = P(T = 9) = \frac{4}{36},$$
$$P(T = 3) = P(T = 11) = \frac{2}{36}, \qquad P(T = 6) = P(T = 8) = \frac{5}{36},$$
$$P(T = 4) = P(T = 10) = \frac{3}{36}, \qquad P(T = 7) = \frac{6}{36}.$$

The distribution is symmetric: $P(T = t) = P(T = 14 - t)$.

Example 2.4 (Children in a Household)**.** Pick a household in the United States uniformly at random. Let X be the number of children in that household. X is a discrete r.v. with support $\{0, 1, 2, \dots\}$. The probability $P(X = x)$ is proportional to the total number of U.S. households with x children.

3.3 Bernoulli and Binomial Distributions

Definition 2.4 (Bernoulli Distribution)**.** An r.v. X has a **Bernoulli distribution** with parameter $p \in (0, 1)$, denoted $X \sim \text{Bern}(p)$, if:

$$P(X = 1) = p \quad \text{and} \quad P(X = 0) = 1 - p.$$

Its PMF is $P(X = x) = p^x(1 - p)^{1-x}$ for $x \in \{0, 1\}$.

Example 2.5 (Canadian Lawyers)**.** Suppose 1 in 200 Canadian adults is a lawyer. Let X indicate whether a randomly selected Canadian adult is a lawyer. Then $X \sim \text{Bern}(1/200)$:

$$P(X = 1) = \frac{1}{200}, \qquad P(X = 0) = \frac{199}{200}.$$

Story 2.1 (Binomial Distribution)**.** Perform n independent Bernoulli trials, each with success probability p. The total number of successes X follows a **Binomial distribution** with parameters n and p, written $X \sim \text{Bin}(n, p)$.

Theorem 2.2 (Binomial PMF)**.** *If $X \sim Bin(n, p)$, then:*

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k \in \{0, 1, \dots, n\}.$$

Proof. Any specific sequence of k successes and $n-k$ failures has probability $p^k(1-p)^{n-k}$ by independence. There are $\binom{n}{k}$ such sequences, each corresponding to a choice of which k trials are successes. □

Theorem 2.3. *If $X \sim Bin(n,p)$ and $q = 1-p$, then $n - X \sim Bin(n,q)$.*

Proof. If X is the number of successes, then $n-X$ is the number of failures. Swapping the roles of success and failure replaces p with q. Alternatively, setting $Y = n - X$:

$$P(Y = k) = P(X = n-k) = \binom{n}{n-k} p^{n-k} q^k = \binom{n}{k} q^k p^{n-k}.$$

□

Corollary 2.1. *If $X \sim Bin(n, 1/2)$ with n even, the distribution of X is symmetric about $n/2$:*

$$P(X = n/2 + j) = P(X = n/2 - j) \quad \text{for all } j \geq 0.$$

3.4 Hypergeometric Distribution

Story 2.2 (Hypergeometric Distribution)**.** An urn contains w white balls and b black balls. We draw n balls without replacement, such that all $\binom{w+b}{n}$ samples are equally likely. The number of white balls X in the sample follows the **Hypergeometric distribution**, written $X \sim \text{HGeom}(w, b, n)$.

Theorem 2.4 (Hypergeometric PMF)**.** *If $X \sim HGeom(w, b, n)$, its PMF is:*

$$P(X = k) = \frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}},$$

for integers k satisfying $\max(0, n-b) \leq k \leq \min(n, w)$.

Example 2.6 (Elk Capture-Recapture)**.** A forest contains N elk. First, m elk are captured, tagged, and released. Later, a sample of n elk are recaptured at random. The number of tagged elk X in the second sample follows:

$$X \sim \text{HGeom}(m, N-m, n).$$

Here, tagged elk are the "white balls" ($w = m$), untagged elk are the "black balls" ($b = N - m$), and the recaptured group is the sample size (n).

Example 2.7 (Aces in a Poker Hand)**.** A 5-card hand is dealt from a well-shuffled 52-card deck. The number of aces follows $\text{HGeom}(4, 48, 5)$. The probability of getting exactly three aces is:

$$P(X = 3) = \frac{\binom{4}{3}\binom{48}{2}}{\binom{52}{5}} = \frac{4 \times 1128}{2{,}598{,}960} \approx 0.00174.$$

Theorem 2.5. *HGeom(w, b, n) and HGeom$(n, w+b-n, w)$ are identical distributions.*

Proof. Classify balls by two sets of tags: Tag 1 (white vs. black) and Tag 2 (sampled vs. not sampled). Both models count items having both tags. Algebraically:

$$\frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}} = \frac{w!\,b!\,n!\,(w+b-n)!}{k!\,(w+b)!\,(w-k)!\,(n-k)!\,(b-n+k)!} = \frac{\binom{n}{k}\binom{w+b-n}{w-k}}{\binom{w+b}{w}}.$$

□

Intuition 2.2 (Binomial vs. Hypergeometric). The Binomial and Hypergeometric distributions both count successes in n trials. The difference lies in whether the trials are independent:

- **Binomial:** Sampling *with* replacement. Trials are independent.
- **Hypergeometric:** Sampling *without* replacement. Trials are dependent because each draw alters the composition of the remaining population.

3.5 Discrete Uniform Distribution

Story 2.3 (Discrete Uniform Distribution)**.** Let C be a finite, non-empty set of numbers. An r.v. X chosen uniformly at random from C follows the **Discrete Uniform distribution**, written $X \sim \text{DUnif}(C)$.

Its PMF is $P(X = x) = \dfrac{1}{|C|}$ for $x \in C$, and for any subset $A \subseteq C$:

$$P(X \in A) = \frac{|A|}{|C|}.$$

Example 2.8 (Random Slips of Paper)**.** A hat contains 100 slips of paper numbered $1, 2, \ldots, 100$. Five slips are drawn.

- **With Replacement:**
  1. Let Y be the number of slips ≥ 80. Then $P(\text{slip } \geq 80) = 21/100 = 0.21$, so $Y \sim \text{Bin}(5, 0.21)$.
  2. The value of the j-th draw is $X_j \sim \text{DUnif}(\{1, 2, \ldots, 100\})$.
  3. The probability that 100 is drawn at least once:

$$1 - P(\text{no } 100) = 1 - \left(\frac{99}{100}\right)^5 \approx 0.0490.$$

- **Without Replacement:**
  1. Let Y be the number of slips ≥ 80. Then $Y \sim \text{HGeom}(21, 79, 5)$.
  2. By symmetry, the j-th draw unconditionally satisfies $Y_j \sim \text{DUnif}(\{1, 2, \ldots, 100\})$.
  3. The probability that 100 is in the sample:

$$P(100 \in \text{sample}) = \frac{\binom{1}{1}\binom{99}{4}}{\binom{100}{5}} = \frac{5}{100} = 0.05.$$

3.6 Cumulative Distribution Functions (CDFs)

Definition 2.5 (Cumulative Distribution Function)**.** The **Cumulative Distribution Function** (CDF) of an r.v. X is:

$$F_X(x) = P(X \leq x).$$

Theorem 2.6 (Properties of Valid CDFs)**.** *A function F is a valid CDF if and only if:*

1. ***Non-decreasing:*** $x_1 \leq x_2 \implies F(x_1) \leq F(x_2)$.
2. ***Right-continuous:*** $\lim_{t \to x^+} F(t) = F(x)$ *for all* $x \in \mathbb{R}$.
3. ***Limits at Infinity:*** $\lim_{x \to -\infty} F(x) = 0$ *and* $\lim_{x \to \infty} F(x) = 1$.

3.7 Functions of Random Variables

Theorem 2.7 (PMF of $g(X)$)**.** *For a discrete r.v. X, the PMF of $Y = g(X)$ is:*

$$P(g(X) = y) = \sum_{x : g(x) = y} P(X = x).$$

If g is one-to-one, this simplifies to $P(g(X) = g(x)) = P(X = x)$.

Example 2.9 (1D Random Walk)**.** A particle starts at 0 and takes n independent steps, each $+1$ or -1 with equal probability $1/2$. Let Y be the particle's position after n steps. Let $X \sim \text{Bin}(n, 1/2)$ be the number of rightward steps. The final position is:

$$Y = X(+1) + (n - X)(-1) = 2X - n.$$

Because $g(x) = 2x - n$ is one-to-one, solving for x gives $x = (n + k)/2$. Thus, for integers $k \in \{-n, -n + 2, \ldots, n\}$:

$$P(Y = k) = P\left(X = \frac{n+k}{2}\right) = \binom{n}{\frac{n+k}{2}} \left(\frac{1}{2}\right)^n.$$

Example 2.10 (Distance from Origin)**.** Continuing the random walk, let $D = |Y|$ be the distance from the origin, assuming n is even. For $k = 0$:

$$P(D = 0) = P(Y = 0) = \binom{n}{n/2} \left(\frac{1}{2}\right)^n.$$

For even integers $k \in \{2, 4, \ldots, n\}$, since $g(y) = |y|$ maps both k and $-k$ to k:

$$P(D = k) = P(Y = k) + P(Y = -k) = 2\binom{n}{\frac{n+k}{2}} \left(\frac{1}{2}\right)^n.$$

Example 2.11 (Maximum of Two Die Rolls)**.** Roll two fair six-sided dice, X and Y. Let $M = \max(X, Y)$. Using the CDF:

$$P(M \leq k) = P(X \leq k, Y \leq k) = P(X \leq k)P(Y \leq k) = \left(\frac{k}{6}\right)^2.$$

The PMF is the difference between consecutive CDF values:

$$P(M = k) = P(M \leq k) - P(M \leq k-1) = \frac{k^2 - (k-1)^2}{36} = \frac{2k-1}{36}.$$

Evaluating for $k = 1, \ldots, 6$ gives:

$$P(M=1) = \frac{1}{36}, \quad P(M=2) = \frac{3}{36}, \quad P(M=3) = \frac{5}{36},$$
$$P(M=4) = \frac{7}{36}, \quad P(M=5) = \frac{9}{36}, \quad P(M=6) = \frac{11}{36}.$$

3.8 Independence of Random Variables

Definition 2.6 (Independence of Two R.V.s)**.** Random variables X and Y are **independent** if:

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y) \quad \text{for all } x, y \in \mathbb{R}.$$

For discrete r.v.s, this is equivalent to:

$$P(X = x, Y = y) = P(X = x)P(Y = y) \quad \text{for all } x, y.$$

Theorem 2.8 (Sum of Independent Binomials)**.** *If $X \sim Bin(n, p)$ and $Y \sim Bin(m, p)$ are independent, then:*

$$X + Y \sim Bin(n + m, p).$$

Proof. Conditioning on X and applying the Law of Total Probability:

$$\begin{aligned}
P(X+Y=k) &= \sum_{j=0}^{k} P(X+Y=k \mid X=j)P(X=j) \\
&= \sum_{j=0}^{k} P(Y=k-j)P(X=j) \quad \text{(by independence)} \\
&= \sum_{j=0}^{k} \binom{m}{k-j} p^{k-j} q^{m-k+j} \binom{n}{j} p^j q^{n-j} \\
&= p^k q^{n+m-k} \sum_{j=0}^{k} \binom{m}{k-j}\binom{n}{j} \\
&= \binom{n+m}{k} p^k q^{n+m-k} \quad \text{(by Vandermonde's Identity).}
\end{aligned}$$

□

Example 2.12 (Matching Pennies)**.** Two players flip fair pennies independently. Let $X \in \{-1, 1\}$ and $Y \in \{-1, 1\}$ indicate Heads for players A and B. Let $Z = XY$ indicate whether they match. X and Y are unconditionally independent, but given $Z = 1$, we know $X = Y$, making them conditionally dependent.

Example 2.13 (Mystery Opponent)**.** You play two tennis games against one of two twins. Against Twin 1, the win probability is 1/2; against Twin 2, it is 3/4. Let Z indicate playing Twin 1 ($P(Z = 1) = 1/2$). Conditional on $Z = 1$, game outcomes X and Y are i.i.d. Bern(1/2). Conditional on $Z = 0$, they are i.i.d. Bern(3/4). Unconditionally, X and Y are dependent: winning the first game makes it more likely that the opponent is the weaker twin, increasing the probability of winning the second.

Chapter 4: Expectation

4.1 Definition of Expectation

Definition 3.1 (Expectation of a Discrete R.V.)**.** The **expected value** (also called expectation or mean) of a discrete r.v. X whose distinct possible values are $x_1, x_2, \ldots$ is:

$$E(X) = \sum_x xP(X = x),$$

provided the sum converges absolutely ($\sum_x |x|P(X = x) < \infty$).

Intuition 3.1. The expected value is the long-run average value of X across many repetitions of an experiment. Geometrically, it is the center of mass of the probability mass function.

Proposition 3.1. *If X and Y are discrete r.v.s with the same distribution, then $E(X) = E(Y)$ (if either side exists). The converse is false: the expected value is just a one-number summary and does not specify the entire distribution.*

Example 3.1 (Fair Six-Sided Die)**.** For a standard die roll X:

$$E(X) = \sum_{j=1}^{6} j\left(\frac{1}{6}\right) = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$

Example 3.2 (Bernoulli Expectation)**.** If $X \sim \text{Bern}(p)$, then:

$$E(X) = 1 \cdot p + 0 \cdot (1-p) = p.$$

4.2 Linearity of Expectation

Theorem 3.1 (Linearity of Expectation)**.** *For any random variables X and Y and any constant c:*

$$E(X+Y) = E(X) + E(Y), \qquad E(cX) = cE(X).$$

This holds whether X and Y are independent or dependent.

Proof. Writing expectations over individual sample space outcomes:

$$E(X) = \sum_{s \in S} X(s)P(\{s\}), \qquad E(Y) = \sum_{s \in S} Y(s)P(\{s\}).$$

Adding the two equations:

$$\begin{aligned} E(X) + E(Y) &= \sum_{s \in S} X(s)P(\{s\}) + \sum_{s \in S} Y(s)P(\{s\}) \\ &= \sum_{s \in S} \big(X(s) + Y(s)\big)P(\{s\}) \\ &= \sum_{s \in S} (X+Y)(s)P(\{s\}) = E(X+Y). \end{aligned}$$

□

Proposition 3.2 (Monotonicity of Expectation)**.** *If $X \geq Y$ with probability 1, then $E(X) \geq E(Y)$.*

Example 3.3 (Binomial Expectation)**.** Let $X \sim \text{Bin}(n, p)$. Express $X = I_1 + \cdots + I_n$, where $I_j \sim \text{Bern}(p)$ indicates success on trial j. By linearity:

$$E(X) = \sum_{j=1}^{n} E(I_j) = \sum_{j=1}^{n} p = np.$$

4.3 Geometric and Negative Binomial Distributions

Story 3.1 (Geometric Distribution)**.** Conduct independent Bernoulli trials with success probability p. Let X be the number of failures *before* the first success. Then $X \sim \text{Geom}(p)$ with support $\{0, 1, 2, \dots\}$.

Theorem 3.2 (Geometric PMF and CDF)**.** *If $X \sim Geom(p)$ and $q = 1 - p$, then:*

$$P(X = k) = q^k p, \quad k \in \{0, 1, 2, \dots\}.$$

Its CDF is:

$$F(x) = 1 - q^{\lfloor x \rfloor + 1}, \quad x \geq 0.$$

Theorem 3.3 (Geometric Expectation)**.** *If $X \sim Geom(p)$, then $E(X) = q/p$.*

Proof. Differentiating both sides of the geometric series $\sum_{k=0}^{\infty} q^k = \dfrac{1}{1-q}$ with respect to q:

$$\sum_{k=1}^{\infty} k q^{k-1} = \frac{1}{(1-q)^2} = \frac{1}{p^2}.$$

Multiplying both sides by pq:

$$E(X) = \sum_{k=0}^{\infty} k q^k p = pq \sum_{k=1}^{\infty} k q^{k-1} = pq \left(\frac{1}{p^2}\right) = \frac{q}{p}.$$

□

Story 3.2 (First Success Distribution)**.** Let Y be the number of trials *up to and including* the first success. Then $Y \sim \text{FS}(p)$, where $Y = X + 1$ with $X \sim \text{Geom}(p)$. Its expectation is:

$$E(Y) = E(X + 1) = \frac{q}{p} + 1 = \frac{1 - p + p}{p} = \frac{1}{p}.$$

Story 3.3 (Negative Binomial Distribution)**.** Let X be the number of failures before the r-th success in independent Bern(p) trials. Then $X \sim \text{NBin}(r, p)$ with PMF:

$$P(X = n) = \binom{n+r-1}{r-1} p^r q^n, \quad n \in \{0, 1, 2, \dots\}.$$

Since $X = X_1 + \cdots + X_r$ where $X_i \overset{\text{i.i.d.}}{\sim} \text{Geom}(p)$, linearity gives:

$$E(X) = \sum_{i=1}^{r} E(X_i) = r\frac{q}{p}.$$

Example 3.4 (Coupon Collector Problem)**.** A toy brand offers n distinct toys. Each purchase yields a uniformly chosen toy. Let N be the total purchases needed to collect all n toys. Decompose $N = N_1 + N_2 + \cdots + N_n$, where N_j is the additional purchases needed to find the j-th new toy. When $j-1$ toys have been found, the probability that the next toy is new is:

$$p_j = \frac{n-(j-1)}{n} = \frac{n-j+1}{n}.$$

Thus $N_j \sim \text{FS}(p_j)$, so $E(N_j) = 1/p_j = \dfrac{n}{n-j+1}$. By linearity:

$$E(N) = \sum_{j=1}^{n} E(N_j) = \sum_{j=1}^{n} \frac{n}{n-j+1} = n\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}\right) \approx n \ln n.$$

Example 3.5 (St. Petersburg Paradox)**.** A fair coin is tossed until the first Heads appears. If Heads appears on toss N, the payoff is $\$2^N$. The payoff is $X = 2^N$, and $P(N = k) = (1/2)^k$. The expected payoff is:

$$E(X) = \sum_{k=1}^{\infty} 2^k P(N = k) = \sum_{k=1}^{\infty} 2^k \left(\frac{1}{2}\right)^k = 1 + 1 + 1 + \cdots = \infty.$$

In practice, resources are finite. If the house caps the game at 40 tosses (a maximum payoff of $2^{40} \approx 1.1 \times 10^{12}$ dollars), the expected payoff becomes:

$$E(X) = \sum_{k=1}^{40} 1 + \sum_{k=41}^{\infty} 0 = 40.$$

4.4 Indicator R.V.s and the Fundamental Bridge

Definition 3.2 (Indicator Random Variable)**.** For any event A, the indicator random variable I_A is 1 if A occurs and 0 otherwise.

Theorem 3.4 (Indicator Properties)**.** *1.* $(I_A)^k = I_A$ *for any positive integer* k*.*

2. $I_{A^c} = 1 - I_A$*.*

3. $I_{A\cap B} = I_A I_B$.
4. $I_{A\cup B} = I_A + I_B - I_A I_B$.

Theorem 3.5 (The Fundamental Bridge)**.** *For any event A:*

$$P(A) = E(I_A).$$

Example 3.6 (Card Matching)**.** A deck of n cards numbered 1 to n is shuffled. A card is a match if its value matches its position in the deck. Let X be the number of matches. Write $X = I_1 + \cdots + I_n$, where I_j indicates that card j is in position j. By symmetry, $P(I_j = 1) = 1/n$. Using the Fundamental Bridge, $E(I_j) = 1/n$. By linearity:

$$E(X) = \sum_{j=1}^{n} E(I_j) = n\left(\frac{1}{n}\right) = 1.$$

The expected number of matches is always 1, regardless of deck size n.

Example 3.7 (Distinct Birthdays and Birthday Matches)**.** In a group of n people, assuming 365 equally likely birthdays:

- **Distinct Birthdays:** Let $X = \sum_{j=1}^{365} I_j$, where I_j indicates that day j is someone's birthday.

$$E(I_j) = P(\text{at least one birthday on day } j) = 1 - \left(\frac{364}{365}\right)^n.$$

  By linearity, $E(X) = 365\left(1 - \left(\frac{364}{365}\right)^n\right)$.
- **Matches:** Let $Y = \sum_{i<j} J_{ij}$, where J_{ij} indicates that persons i and j share a birthday. There are $\binom{n}{2}$ pairs, each with match probability 1/365. By linearity:

$$E(Y) = \frac{\binom{n}{2}}{365}.$$

Example 3.8 (Putnam Permutations Problem)**.** A permutation $a_1, \ldots, a_n$ of $1, \ldots, n$ has a local maximum at position j if a_j exceeds its neighbors ($a_j > a_{j-1}$ and $a_j > a_{j+1}$ for interior positions; $a_1 > a_2$ for $j = 1$; $a_n > a_{n-1}$ for $j = n$). Let I_j indicate a local maximum at position j.

- For $2 \le j \le n-1$: a_j must be the largest of 3 elements. By symmetry, $E(I_j) = 1/3$.
- For $j = 1$ and $j = n$: a_j must be the largest of 2 elements. By symmetry, $E(I_j) = 1/2$.

By linearity:

$$E\left(\sum_{j=1}^{n} I_j\right) = 2\left(\frac{1}{2}\right) + (n-2)\left(\frac{1}{3}\right) = 1 + \frac{n-2}{3} = \frac{n+1}{3}.$$

Example 3.9 (Negative Hypergeometric Expectation)**.** An urn contains w white balls and b black balls. Balls are drawn without replacement until r white balls are obtained. Let $X \sim \text{NHGeom}(w, b, r)$ be the number of black balls drawn. Label the black balls $1, \ldots, b$. Let I_j indicate that black ball j is drawn before the r-th white ball. By symmetry, in any ordered subset consisting of black ball j and the w white balls, each position is equally likely. The event occurs if black ball j is before the r-th white ball (i.e., in the first r positions out of $w+1$). Thus $P(I_j = 1) = \dfrac{r}{w+1}$. By linearity:

$$E(X) = \sum_{j=1}^{b} E(I_j) = \frac{rb}{w+1}.$$

Theorem 3.6 (Expectation via Survival Function)**.** *For a nonnegative integer-valued r.v. X:*

$$E(X) = \sum_{n=0}^{\infty} P(X > n).$$

Proof. Represent X as a sum of indicators: $X = \sum_{k=1}^{\infty} I(X \geq k)$. Taking expectations:

$$E(X) = \sum_{k=1}^{\infty} E[I(X \geq k)] = \sum_{k=1}^{\infty} P(X \geq k) = \sum_{n=0}^{\infty} P(X > n).$$

□

4.5 Law of the Unconscious Statistician (LOTUS)

Theorem 3.7 (LOTUS)**.** *If X is a discrete r.v. and $g : \mathbb{R} \to \mathbb{R}$, then:*

$$E(g(X)) = \sum_{x} g(x) P(X = x).$$

Proof. Writing the expectation over the sample space outcomes:

$$E(g(X)) = \sum_{s \in S} g(X(s)) P(\{s\}).$$

Partitioning the sample space by the values x taken by X:

$$E(g(X)) = \sum_{x} \sum_{s: X(s) = x} g(X(s)) P(\{s\}) = \sum_{x} g(x) \sum_{s: X(s) = x} P(\{s\}) = \sum_{x} g(x) P(X = x).$$

□

4.6 Variance

Definition 3.3 (Variance and Standard Deviation)**.** The **variance** of an r.v. X is:

$$\mathrm{Var}(X) = E\big[(X - EX)^2\big].$$

The standard deviation is $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$.

Theorem 3.8 (Alternative Formula for Variance)**.**

$$\mathrm{Var}(X) = E(X^2) - (EX)^2.$$

Proof. Let $\mu = EX$. Expanding the quadratic:

$$\begin{aligned} E\big[(X-\mu)^2\big] &= E(X^2 - 2\mu X + \mu^2) \\ &= E(X^2) - 2\mu E(X) + \mu^2 \\ &= E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2. \end{aligned}$$

□

Theorem 3.9 (Properties of Variance)**.** *1.* $\mathrm{Var}(X + c) = \mathrm{Var}(X)$ *for any constant* c*.*
2. $\mathrm{Var}(cX) = c^2 \mathrm{Var}(X)$ *for any constant* c*.*
3. If X *and* Y *are independent, then* $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$*.*
4. $\mathrm{Var}(X) \geq 0$*, with* $\mathrm{Var}(X) = 0$ *if and only if* $P(X = c) = 1$ *for some constant* c*.*

Example 3.10 (Geometric Variance)**.** Let $X \sim \mathrm{Geom}(p)$. Differentiating the geometric series $\sum_{k=0}^{\infty} q^k = \dfrac{1}{1-q}$ yields $\sum_{k=1}^{\infty} kq^{k-1} = \dfrac{1}{(1-q)^2}$. Multiplying by q:

$$\sum_{k=1}^{\infty} kq^k = \frac{q}{(1-q)^2}.$$

Differentiating again with respect to q:

$$\sum_{k=1}^{\infty} k^2 q^{k-1} = \frac{(1-q)^2 - q(2(1-q)(-1))}{(1-q)^4} = \frac{(1-q) + 2q}{(1-q)^3} = \frac{1+q}{(1-q)^3}.$$

Multiplying by pq:

$$E(X^2) = \sum_{k=1}^{\infty} k^2 q^k p = pq\left(\frac{1+q}{(1-q)^3}\right) = \frac{q(1+q)}{p^2}.$$

Thus:

$$\mathrm{Var}(X) = E(X^2) - (EX)^2 = \frac{q(1+q)}{p^2} - \frac{q^2}{p^2} = \frac{q}{p^2}.$$

Example 3.11 (Binomial Variance)**.** Let $X \sim \text{Bin}(n,p)$, so $X = \sum_{j=1}^{n} I_j$ with I_j i.i.d. Bern(p). For each indicator:

$$\text{Var}(I_j) = E(I_j^2) - (E(I_j))^2 = p - p^2 = p(1-p).$$

Because the trials are independent, variances add directly:

$$\text{Var}(X) = \sum_{j=1}^{n} \text{Var}(I_j) = np(1-p).$$

4.7 Poisson Distribution

Definition 3.4 (Poisson Distribution)**.** An r.v. X has the **Poisson distribution** with parameter $\lambda > 0$, written $X \sim \text{Pois}(\lambda)$, if its PMF is:

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k \in \{0, 1, 2, \dots\}.$$

Theorem 3.10 (Poisson Mean and Variance)**.** *If $X \sim \text{Pois}(\lambda)$, then $E(X) = \lambda$ and $\text{Var}(X) = \lambda$.*

Proof. For the expectation:

$$E(X) = \sum_{k=0}^{\infty} k\frac{e^{-\lambda}\lambda^k}{k!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} e^{\lambda} = \lambda.$$

To find $E(X^2)$, differentiate the series $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$ twice with respect to λ:

$$\frac{d}{d\lambda}\left(\sum_{k=0}^{\infty} \frac{\lambda^k}{k!}\right) = \sum_{k=1}^{\infty} k\frac{\lambda^{k-1}}{k!} = e^{\lambda} \implies \sum_{k=1}^{\infty} k\frac{\lambda^k}{k!} = \lambda e^{\lambda}.$$

Differentiating a second time:

$$\sum_{k=1}^{\infty} k^2\frac{\lambda^{k-1}}{k!} = \frac{d}{d\lambda}(\lambda e^{\lambda}) = e^{\lambda}(1+\lambda) \implies \sum_{k=1}^{\infty} k^2\frac{\lambda^k}{k!} = \lambda(1+\lambda)e^{\lambda}.$$

Multiplying by $e^{-\lambda}$:

$$E(X^2) = e^{-\lambda} \sum_{k=0}^{\infty} k^2\frac{\lambda^k}{k!} = \lambda(1+\lambda).$$

Therefore:

$$\text{Var}(X) = E(X^2) - (EX)^2 = \lambda(1+\lambda) - \lambda^2 = \lambda.$$

□

Approximation 3.1 (Poisson Paradigm). Let $A_1, \ldots, A_n$ be events with $p_j = P(A_j)$, where n is large, the p_j are small, and the A_j are independent or weakly dependent. Let

$$X = \sum_{j=1}^{n} I(A_j)$$

count how many of the A_j occur. Then X is approximately distributed as Pois(λ), where

$$\lambda = \sum_{j=1}^{n} p_j.$$

Example 3.12 (Balls in Boxes)**.** Place k distinguishable balls randomly into n distinguishable boxes.

1. Let I_j indicate that box j is empty. Then $P(I_j = 1) = (1 - 1/n)^k$. By linearity, the expected number of empty boxes is:
$$E(X) = n\left(1 - \frac{1}{n}\right)^k.$$
2. The exact probability that at least one box is empty, by Inclusion-Exclusion, is:
$$P\left(\bigcup_{j=1}^{n} A_j\right) = \sum_{j=1}^{n-1} (-1)^{j+1} \binom{n}{j} \left(1 - \frac{j}{n}\right)^k.$$
3. For $n = 1000$ and $k = 5806$, $E(X) \approx 3$. Using the Poisson approximation $X \sim \text{Pois}(3)$:
$$P(X \geq 1) = 1 - P(X = 0) \approx 1 - e^{-3} \approx 1 - \frac{1}{20} = 0.95.$$

Example 3.13 (Birth-Minute and Birth-Hour Matches)**.** Among $n = 1600$ students:

1. **Same Day, Hour, and Minute:** There are $c = 365 \times 24 \times 60 = 525{,}600$ possible slots. The expected number of coincident pairs is:
$$\lambda_1 = \binom{n}{2}\frac{1}{c} = \binom{1600}{2}\frac{1}{525{,}600} \approx 2.4338.$$
The probability of at least one pair is $1 - e^{-\lambda_1} \approx 1 - e^{-2.4338} \approx 0.9122$.
2. **Four People with Same Day and Hour:** There are $b = 365 \times 24 = 8760$ day-hours.
   - *Method 1 (by 4-tuples):* Expected number of matching 4-tuples is:
   $$\lambda_2 = \binom{n}{4}\frac{1}{b^3} = \binom{1600}{4}\frac{1}{8760^3} \approx 0.405.$$
   The probability of at least one match is approximately $1 - e^{-0.405} \approx 0.333$.
   - *Method 2 (by day-hours):* Let I_j indicate ≥ 4 births in the j-th slot. With $p = 1/b$ and $q = 1 - p$:
   $$P(I_j = 1) = 1 - \sum_{k=0}^{3} \binom{n}{k} p^k q^{n-k}.$$
   The expected number of crowded slots is $\lambda_3 = bP(I_1 = 1) \approx 0.350$, yielding $1 - e^{-0.350} \approx 0.295$.

4.8 Connections Between Poisson and Binomial

Theorem 3.11 (Theorem 4.8.1: Sum of Independent Poissons)**.** *If* $X \sim \text{Pois}(\lambda_1)$ *and* $Y \sim \text{Pois}(\lambda_2)$ *are independent, then:*

$$X + Y \sim \text{Pois}(\lambda_1 + \lambda_2).$$

Proof. Applying the Law of Total Probability by conditioning on X:

$$\begin{aligned}
P(X+Y=k) &= \sum_{j=0}^{k} P(X+Y=k \mid X=j)P(X=j) \\
&= \sum_{j=0}^{k} P(Y=k-j)P(X=j) \\
&= \sum_{j=0}^{k} \frac{e^{-\lambda_2}\lambda_2^{k-j}}{(k-j)!}\frac{e^{-\lambda_1}\lambda_1^{j}}{j!} \\
&= \frac{e^{-(\lambda_1+\lambda_2)}}{k!}\sum_{j=0}^{k}\binom{k}{j}\lambda_1^j\lambda_2^{k-j} \\
&= \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^k}{k!}.
\end{aligned}$$

□

Theorem 3.12 (Theorem 4.8.2: Poisson Component Given Sum)**.** *If* $X \sim \text{Pois}(\lambda_1)$ *and* $Y \sim \text{Pois}(\lambda_2)$ *are independent, then:*

$$X \mid (X+Y=n) \sim Bin\left(n, \frac{\lambda_1}{\lambda_1+\lambda_2}\right).$$

Proof. Using Bayes' Rule and Theorem 4.8.1:

$$\begin{aligned}
P(X=k \mid X+Y=n) &= \frac{P(X+Y=n \mid X=k)P(X=k)}{P(X+Y=n)} \\
&= \frac{P(Y=n-k)P(X=k)}{P(X+Y=n)} \\
&= \frac{\left(\frac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!}\right)\left(\frac{e^{-\lambda_1}\lambda_1^{k}}{k!}\right)}{\frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}} \\
&= \binom{n}{k}\left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k\left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-k}.
\end{aligned}$$

□

Theorem 3.13 (Theorem 4.8.3: Poisson Approximation to Binomial)**.** *If $X \sim Bin(n,p)$ with $n \to \infty$ and $p \to 0$ such that $\lambda = np$ remains fixed (or $np \to \lambda$), then for any integer $k \geq 0$:*

$$P(X = k) \to \frac{e^{-\lambda}\lambda^k}{k!}.$$

Proof. Writing the Binomial PMF in terms of $\lambda = np$:

$$\begin{aligned} P(X = k) &= \binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k}{k!}\left(\frac{n(n-1)\dots(n-k+1)}{n^k}\right)\left(1 - \frac{\lambda}{n}\right)^n\left(1 - \frac{\lambda}{n}\right)^{-k}. \end{aligned}$$

Taking the limit as $n \to \infty$:

- $\dfrac{n(n-1)\dots(n-k+1)}{n^k} \to 1.$
- $\left(1 - \dfrac{\lambda}{n}\right)^n \to e^{-\lambda}.$
- $\left(1 - \dfrac{\lambda}{n}\right)^{-k} \to 1.$

Combining these yields $P(X = k) \to \dfrac{e^{-\lambda}\lambda^k}{k!}$. □

Example 3.14 (Example 4.8.4: Website Visitors)**.** A website receives traffic from $n = 1{,}000{,}000$ independent potential visitors each day, each with visit probability $p = 2 \times 10^{-6}$. The parameter is $\lambda = np = 2$. Approximating the count by $X \sim \text{Pois}(2)$, the probability of receiving at least 3 visitors is:

$$\begin{aligned} P(X \geq 3) &= 1 - P(X \leq 2) \\ &= 1 - e^{-2}\left(\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!}\right) \\ &= 1 - e^{-2}(1 + 2 + 2) = 1 - 5e^{-2} \approx 0.3233. \end{aligned}$$

4.9 *Using Probability and Expectation to Prove Existence

Intuition 3.2 (The Probabilistic Method). To prove that a deterministic mathematical object with a certain property exists, define a suitable probability space and show that a randomly chosen object has the desired property with positive probability. In terms of expectation:

- If $E(X) = c$, then there must exist at least one outcome where $X \geq c$, and at least one outcome where $X \leq c$.
- If X takes integer values and $E(X) > c$, there exists an outcome with $X \geq \lfloor c \rfloor + 1$.

Example 3.15 (Putnam Matchings Problem)**.** Let $\mathcal{F}$ be a family of subsets of $\{1, 2, \dots, n\}$ such that no subset in $\mathcal{F}$ is contained in another. Let s_k denote the number of subsets of size k in

$\mathcal{F}$. Let a permutation of $1, \ldots, n$ be chosen uniformly at random. Let A be the set of elements in the permutation that precede the first element not in some $F \in \mathcal{F}$. By establishing indicators and using the fact that at most one prefix can belong to $\mathcal{F}$, the probabilistic method gives the Lubell-Yamamoto-Meshalkin (LYM) inequality:

$$\sum_{k=1}^{n} \frac{s_k}{\binom{n}{k}} \leq 1.$$

Theorem 3.14 (Ramsey Lower Bound)**.** *The Ramsey number $R(k,k)$ is the minimum integer n such that every 2-coloring of the edges of the complete graph K_n contains a monochromatic K_k. If $\binom{n}{k} 2^{1-\binom{k}{2}} < 1$, then $R(k,k) > n$. In particular, $R(k,k) > 2^{k/2}$ for $k \geq 3$.*

Proof. Color each edge of K_n independently red or blue with probability $1/2$. For any set S of k vertices, let I_S indicate that the induced subgraph on S is monochromatic. Since there are $\binom{k}{2}$ edges in S:

$$P(I_S = 1) = 2 \cdot \left(\frac{1}{2}\right)^{\binom{k}{2}} = 2^{1-\binom{k}{2}}.$$

Let $X = \sum_{|S|=k} I_S$ be the total number of monochromatic K_k's. By linearity:

$$E(X) = \binom{n}{k} 2^{1-\binom{k}{2}}.$$

If $E(X) < 1$, then $P(X = 0) > 0$. Hence, there exists a 2-coloring of K_n with no monochromatic K_k, which implies $R(k,k) > n$. □